

Fintech Payment Analytics Dashboard

Analysis of 6.3M Transactions | Fraud Detection & Pattern Intelligence

Total Transactions

6.36M

Total Fraud Cases

8197

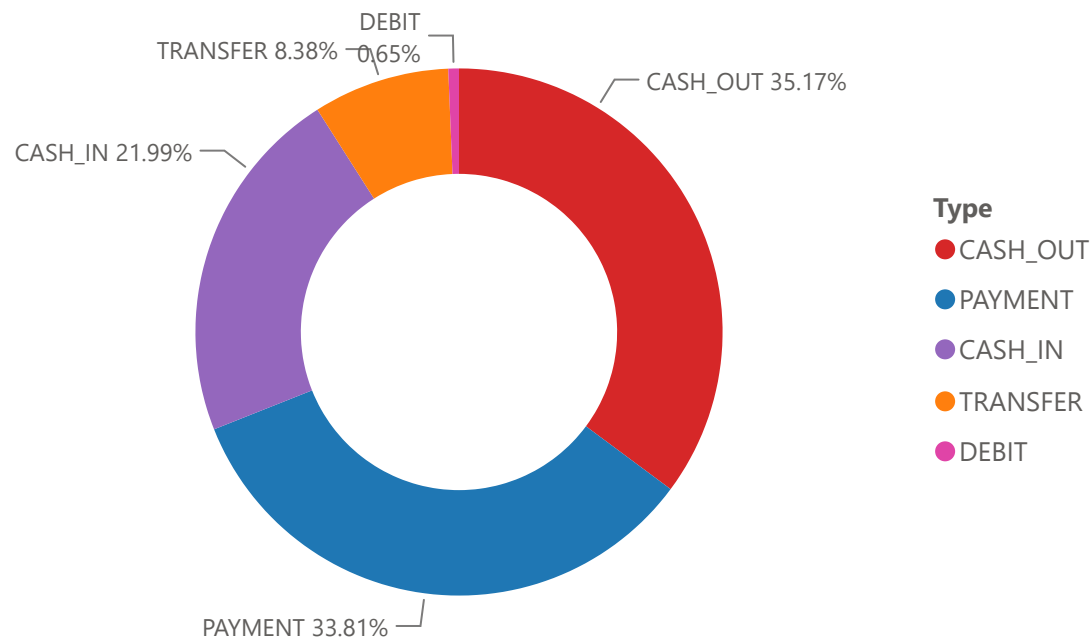
Fraud Rate (%)

0.1288

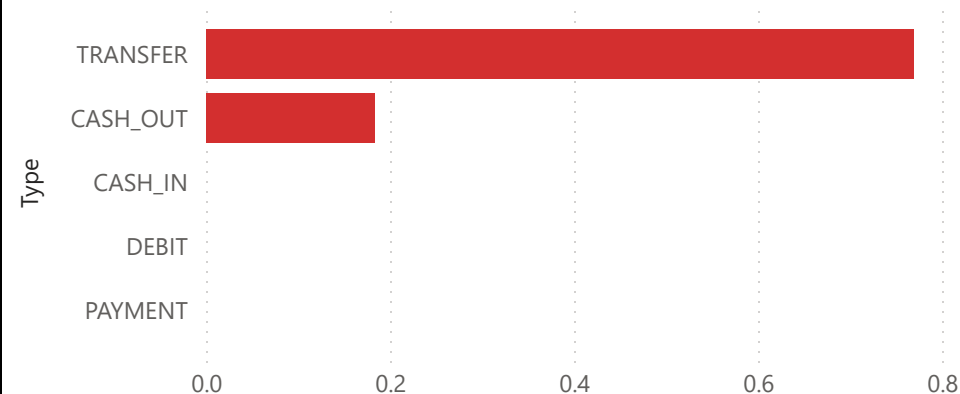
Total Fraud Value

\$12.06bn

Transaction Volume by Type



Fraud Rate by Transaction Type (%)



Key Findings: Analyzed 6.36M payment transactions identifying 8,197 fraud cases with a 0.13% fraud rate representing \$12.06bn in total fraud value. Fraud is exclusively concentrated in TRANSFER and CASH_OUT transaction types - PAYMENT, DEBIT, and CASH_IN recorded zero fraud cases across 5.79M transactions. TRANSFER transactions carry the highest fraud rate at 0.77% despite representing only 8.38% of total transaction volume.

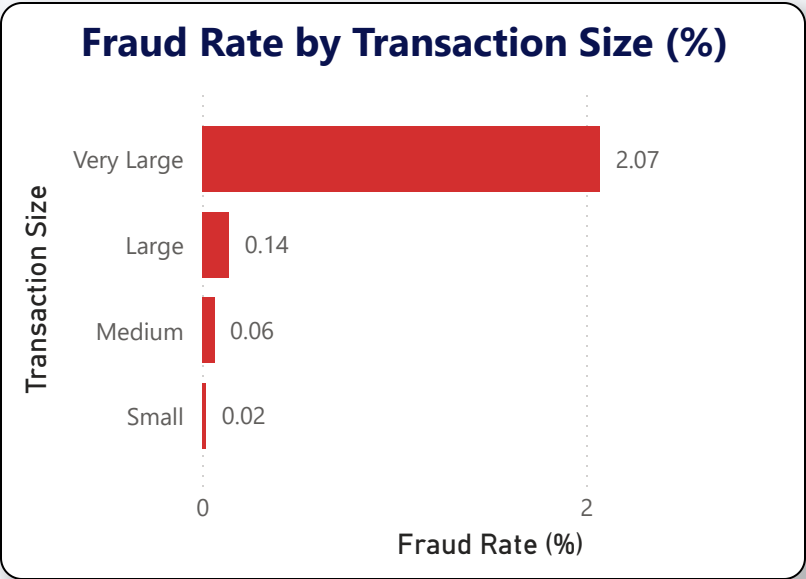
Fraud Pattern Deep Dive

Top 30 Highest Risk Transaction Patterns							
Type	Transaction Size	Time	Total Txns	Fraud Count	Fraud Rate (%)	Fraud Value	Avg Amount
TRANSFER	Very Large	Night	1974	341	17.27	1,270.0776M	25,69,019.90
TRANSFER	Medium	Night	1680	174	10.36	8.7618M	55,441.04
TRANSFER	Large	Night	5972	449	7.52	199.0132M	4,00,894.81
CASH_OUT	Large	Night	11714	451	3.85	199.5621M	2,47,498.36
CASH_OUT	Medium	Night	5573	174	3.12	8.7618M	55,297.26
TRANSFER	Medium	Morning	8500	186	2.19	9.0707M	54,690.78
TRANSFER	Small	Afternoon	2417	32	1.32	0.1269M	5,048.12
TRANSFER	Small	Evening	2505	33	1.32	0.1710M	4,971.51
TRANSFER	Very Large	Evening	32255	342	1.06	1,396.9829M	21,85,562.61
TRANSFER	Very Large	Morning	34983	338	0.97	1,246.4466M	22,52,874.82
TRANSFER	Medium	Afternoon	22015	188	0.85	9.4519M	55,240.39

Key Findings:

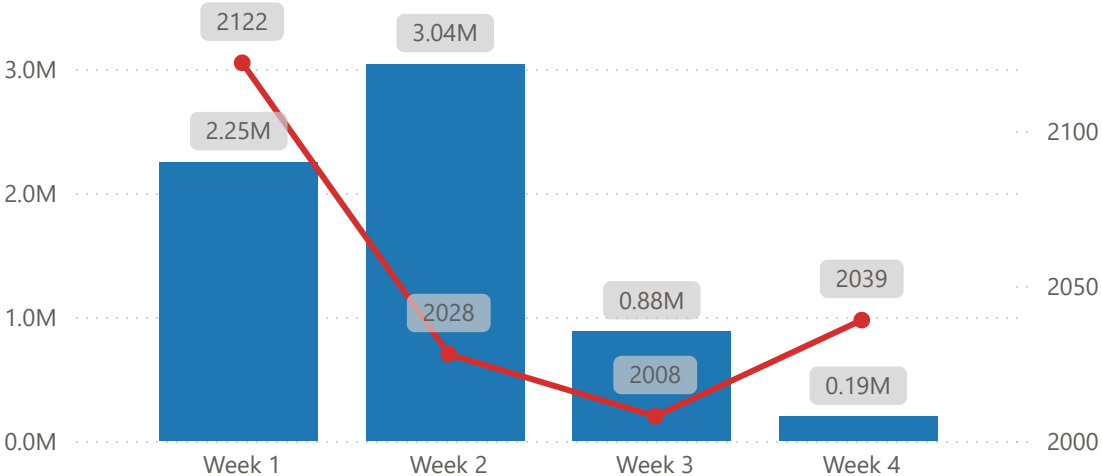
Very Large transactions carry the highest fraud rate at 2.07%, 16x higher than Small transactions at 0.02% - and account for \$10.46bn in total fraud value despite representing only 2.05% of transaction volume. TRANSFER transactions during Night hours exhibit the highest fraud rate at 17.27% - identifying nighttime high-value transfers as the single most dangerous transaction pattern. Fraud activity is entirely absent in PAYMENT, DEBIT, and CASH_IN transactions - confirming fraud risk is structurally concentrated in two transaction types.

Fraud Rate Heatmap - Type vs Time of Day				
Type	Afternoon	Evening	Morning	Night
TRANSFER	3.0679	3.4346	3.9952	35.1501
PAYMENT				0.0000
CASH_OUT	0.2442	0.2657	0.6037	6.9723
CASH_IN	0.0000	0.0000	0.0000	0.0000

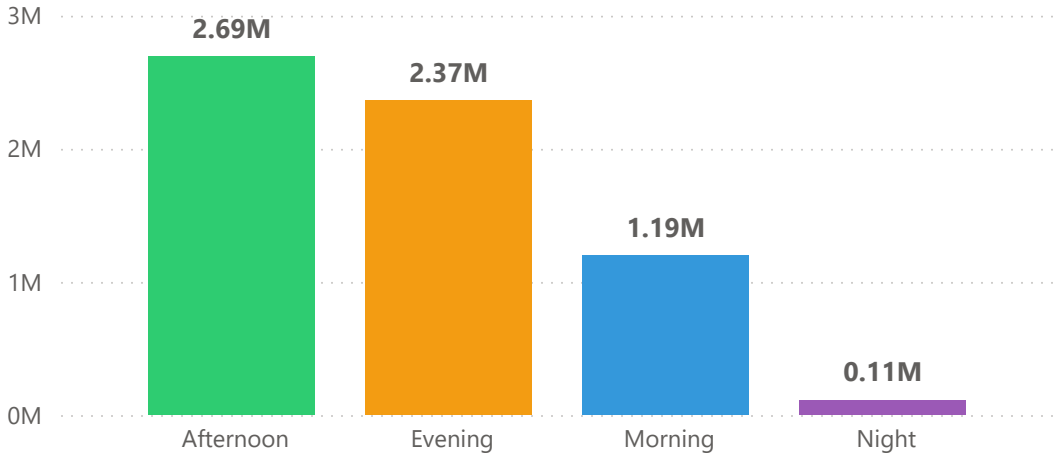


Transaction Volume & Trend Analysis

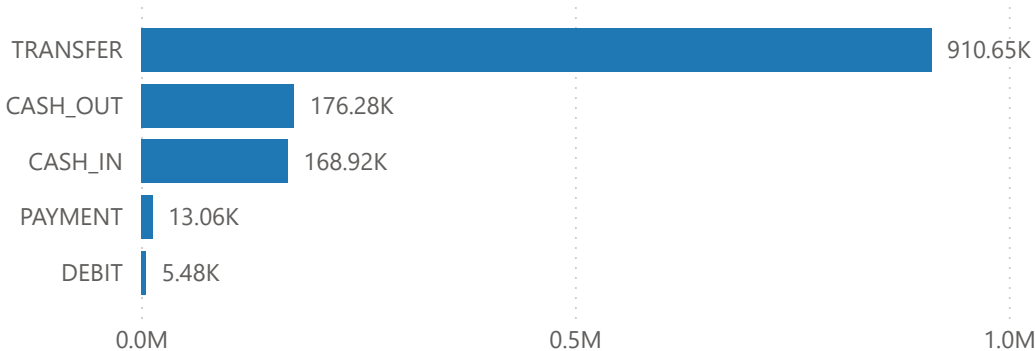
Weekly Transaction Volume vs Fraud Count



Transaction Volume by Time of Day



Average Transaction Value by Type



Key Findings:

Transaction volume peaked in Week 2 with 3.04M transactions before declining sharply to 0.19M in Week 4 - however fraud rate increased to 1.05% in Week 4 suggesting fraud concentration rises as volume drops. Afternoon recorded the highest transaction volume at 2.69M while Night had the lowest at 0.11M - yet Night transactions carry the highest fraud rate at 1.77%, 23x higher than Afternoon at 0.08%. TRANSFER transactions have the highest average transaction value at \$910,647, 52x higher than PAYMENT at \$13,058 - indicating high-value transfers represent the greatest financial exposure.

Fraud Detection System Analysis

Fraud Correctly Flagged

16

Fraud Missed by System

8181

False Positive

0

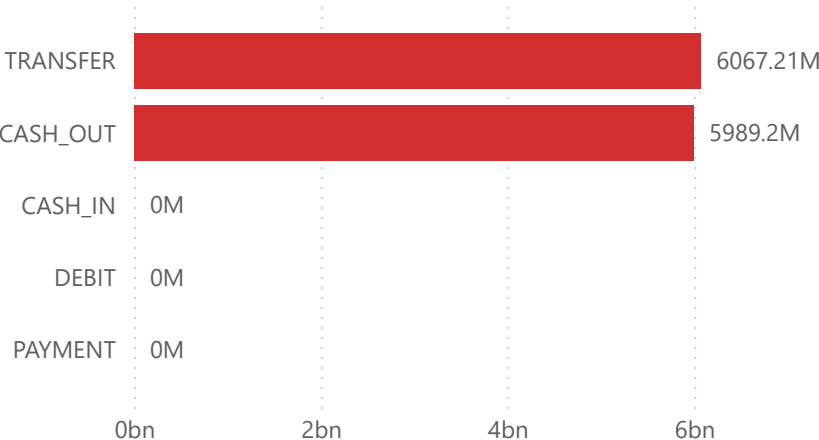
System Detection Rate

0.1952%

Fraud Detection System Performance

System Flag	Actual Flag	Transactions	Total Transaction
2	1	6.36M	11,44,392.94M

Total Fraud Value by Transaction Type



Critical Findings: The existing fraud detection system correctly identified only 16 of 8,197 fraudulent transactions - a 0.20% detection rate - leaving 8,181 fraud cases worth \$11.98bn undetected. Zero false positives were generated indicating high system precision but critically low coverage. Fraud exposure across TRANSFER and CASH_OUT transactions totals \$12.06bn - \$6.07bn and \$5.99bn respectively. Deploying machine learning models targeting TRANSFER and CASH_OUT transactions during Night and Morning hours - the highest risk segments identified - could substantially improve detection coverage from 0.20% toward industry standard rates of 80%+."